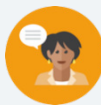


1.2 Evaluating Functions in Function Notation

Lesson Introduction

 Benchmark	MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.		 Learning Target	I can evaluate a function for a specific input value of the domain.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or a mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.		MA.912.F.3.3 Solve mathematical and real-world problems involving functions that have been combined using arithmetic operations.	
 Essential Questions	- How can I determine if a relation is a function? - What is the difference between evaluating $f(5)$ and $f(x) = 5$?			
 Lesson Narrative	In this lesson, students are introduced to the concept of a function. They will learn the key vocabulary associated with functions. Additionally, students will apply this knowledge to evaluate functions provided in the form of algebraic expression, graph, or mapping diagram.			
 Component Summary	1.2.1 Warm-Up (~5 min.)	Food science career exploration activity (<i>SE p. 10</i>)	1.2.4 Guided Instruction (15 min.)	Evaluating functions activity (<i>SE p. 12</i>)
	1.2.2 Collaboration (5 min.)	Introduction to the concept of a function, represented in multiple ways (<i>SE p. 11</i>)	1.2.5 Try It (5 min.)	Evaluating functions provided in multiple forms practice (<i>SE p. 14</i>)
	1.2.3 Exploration (5 min.)	Exploration of the different graphs of functions, noticing basic behaviors of each one (<i>SE p. 11</i>)	1.2.6 Your Turn! (5 min.)	Evaluating functions provided in multiple forms independent practice (<i>SE p. 15</i>)
	1.2.7 Wrap-Up (~5 min.)	Ticket Out the Door on evaluating functions. (<i>SE p. 15</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Relation - Function <u>Additional Terms:</u> - Function Notation - Input - Output		(Optional) Chart Paper for Anchor Chart	Warm-Up 1.2.1 contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions - Students may not accurately differentiate between evaluating $f(5)$ vs $f(x) = 5$. - Students may make calculation errors in the evaluation. - Students may switch the input and output values on a graph. - Students may think repeating output values makes a relation not a function.	Things to Consider This lesson contains a lot of new vocabulary and concepts. Consider creating a foldable or guided notes document for students to use as a later reference with examples of different representations.
	Literacy and Language Routines Compare and Connect 1.2.4 Guided Instruction provides students with an opportunity to receive guided instruction on how to evaluate functions, represented in several forms. Consider leveraging the Compare and Connect protocol during this activity. For each question set, model one question by thinking out loud and questioning why it might be done this way, prompting students to reflect and respond prior to being released to try on their own.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency 1.2.5 Try It has students working to practice evaluating functions in multiple forms on their own. Upon completion, they will have their work checked by their partner. Each partner will provide the other with feedback about their final answer based on the input values and calculations. Students will use this feedback to improve their efficiency when performing calculations.
	1.2.5 Try It provides students with an opportunity to practice independently on evaluating functions. Student performance in this activity can drive potential differentiation in 1.2.6 Your Turn! if students may need additional support. Consider the use of an anchor chart or alternative when introducing this concept. Students may not be familiar with function language. Creating a visual aid for students to have can assist students with knowing what to do in a situation where $f(5)$ is being presented vs where $f(x) = 5$ is being presented. It can also assist students with knowing how to read and interpret functions. The 1.2.7 Wrap-Up for this lesson is a Ticket Out the Door that will provide formative feedback on how well any given student is performing towards mastery of the concepts in this lesson.	
 Differentiate Instruction		
Lesson Notes:		

1.2.1 Warm-Up: Food and Research Career

Component Overview: Play the Career Profile video on a Food and Research Career for the entire class. After students watch the video (1:22 minutes), prompt them to reflect on the profession by answering the two questions independently.



1.2.1 Warm-Up: Food and Research Career

Richard works in Research and Development for Kellogg's. One of the projects he is most proud of involves developing a line of organics products and no added sugar products. Richard also points out that working with a team provides an exciting variety of experiences that is crucial and rewarding to the work of the company.

1. How could mathematics help in the food science industry?
Answers will vary.
2. Why is it important to have teamwork skills when working for a company?
Answers will vary.



Have you ever heard of the food science industry? What would it be like if the person doing this job did not have teamwork skills?

1.2.2 Collaboration: Functions

Component Overview: Students will be introduced to the concept of a function, represented in multiple ways. For this activity, begin by reviewing the formal definitions of a relation and a function and review the three examples that will be used for this activity.



1.2.2 Collaboration: Functions

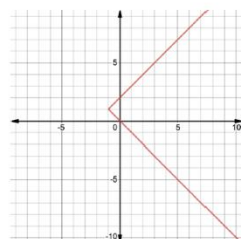
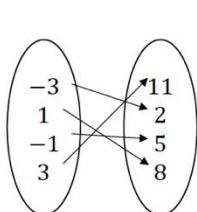


A **relation** is a set of input-output pairs.



A **function** is a mathematical relation for which each element in the domain corresponds to exactly one element in the range.

Three different relations are shown, with each represented in a different way.



x	y
-4	2
7	0
-4	-4
0	-6

- With your partner, write an explanation identifying which of the relations above are functions.
The first relation is a function. Each input has exactly one output.
The second relation is not a function. The x-values (input) have two different y-values (outputs).
The third relation is not a function. The x-values (input) have two different y-values (outputs).



Understanding the difference between a relation and a function is taught in grade 8, MA.8.F.1.1. This Collaboration is designed to help students recall the definition of a relation and a function. The benchmark, MA.912.F.1.2, does not require a student to distinguish between a relation and a function. Reinforcing the vocabulary can be done throughout the course.

Consider giving students an example of a relation that is not a function, and then transition to an example where the relation is a function.



- How did you use the definitions to determine if each relation was a function?
- Were there any representations that were easier to work with?

1.2.3 Exploration: Graphs of Functions

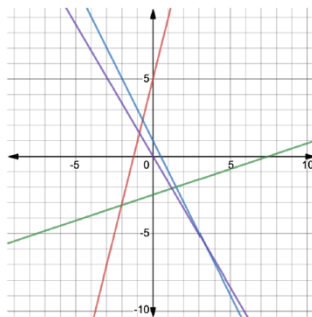
Component Overview: Students will explore the different graphs of functions, noticing basic behaviors of each one. For this activity, begin by introducing students to the context of the question, including matching each line with its corresponding equation using the matching colors. For questions 1 – 4, release students to explore these questions and answer them with a partner after the explanation of the context. Bring the class back together to review their answers to each question.



1.2.3 Exploration: Graphs of Functions

The equations and graphs of four linear functions are shown.

$$\begin{aligned} y &= 4x + 5 \\ y &= -2x + 1 \\ y &= \frac{1}{3}x - 2.5 \\ y &= -1.7x \end{aligned}$$



- Write the equations of the lines that decrease from left to right.
 $y = -2x + 1$
 $y = -1.7x$
- Write the equations of the lines that increase from left to right.
 $y = 4x + 5$
 $y = \frac{1}{3}x - 2.5$
- What do the equations of the lines that increase from left to right have in common?
Answers will vary. The slopes of the lines are positive. The coefficients of x are positive.
- How are the equations of the lines that decrease from left to right different from the equations of the lines that increase from left to right?
Answers will vary. The slopes of the lines are negative. The coefficients of x are negative.



The graph of each line and its corresponding equation are matching by using the same color. Draw attention to how students can identify the line that matches each equation. If students have difficulty seeing the colors, consider labeling each equation and then labeling their corresponding graph on the coordinate grid.



- Why do you think that all these questions use the phrase from “left to right?”
- Why do you think that it is important to have a standard way of reading a graph?



Allow students to share out their answers for each question. Highlight any misconceptions that are heard such as the way the graph was read. For questions 3 and 4, students may not use the vocabulary of coefficients or slope when indicating negative or positive. Clarify which value is negative or positive in both.

1.2.4 Guided Instruction: Evaluating Functions

Component Overview: Students will receive guided instruction on how to evaluate functions. For this activity, begin by providing students with an introduction into the vocabulary of a function including function notation, name of function, input of function, and output of function. Use the narrative information prior to question 1 to assist with providing this instruction.

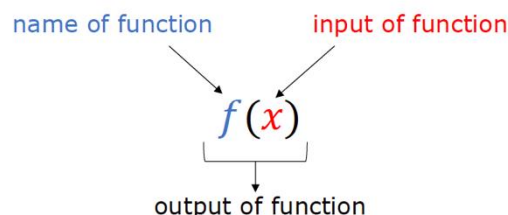


1.2.4 Guided Instruction: Evaluating Functions

Each of the functions in the Exploration starts with “ $y =$.” If asked to use the function y from the list of functions to find a specific value, it would not be clear which of the four functions to use.

Since mathematicians often work with multiple functions at a time, **function notation** is used to name specific functions.

The notation $y = f(x)$ defines a function named f . This indicates “ f is a function of x .” The letter x represents the input value, or independent variable. The letter y , or $f(x)$, represents the output value, or dependent variable. The notation $f(x)$ is read, “ f of x .”



A function can be named using any letter. For example, the previous functions can be renamed as shown:

$$f(x) = 4x + 5 \qquad g(x) = -2x + 1$$

$$h(x) = \frac{1}{3}x - 2.5 \qquad c(x) = -1.7x$$

Now, if asked to use the function h to find a specified value of x , it would be clear which function to use.



Consider the use of an anchor chart or alternative when introducing this concept. Students may not be familiar with function language. Creating a visual aid for students can assist them when presented with a situation such as $f(5)$ vs. $f(x) = 5$. It can also assist students in reading and interpreting the functions.



For the functions provided, model the first one for students on how to read it properly. Have students take turns reading the additional three to practice how to read each function correctly. Use these functions to reinforce the definition of a function that was introduced earlier in the lesson.

1.2.4 Guided Instruction: Evaluating Functions (continued)

Functions are evaluated in the same way equations are evaluated when given a value for a variable. Consider the function $f(w) = -7w^2 - 10w + 3$.

- Find the value $f(10)$.

$$f(w) = -7w^2 - 10w + 3$$

$$f(10) = -7(10)^2 - 10(10) + 3$$

$$f(10) = -797$$

When the input is 10, the output of the function is -797 .

Using function notation, this is written $f(10) = -797$.

- Find the value $f(-3)$.

$$f(-3) = -30$$

This same function $f(w)$ can be represented in other ways.

Shown here are a table of values, a graph, and a mapping diagram. The function can be evaluated from each of these representations.

Table of Values

w	$f(w)$
-15	-1422
-10	-597
-1	6
2	-45

- Find $f(-10)$.

$$f(-10) = -597$$

- If $f(w) = -45$, what is the value of w ?
 $w = 2$



Prior to modeling question 1, review the function that is provided for the context for questions 1 and 2. Remind students of how this function is read, and what it means for an input value. Take a moment to remind students to take their time when evaluating to avoid making a calculation error with exponents or negative signs.

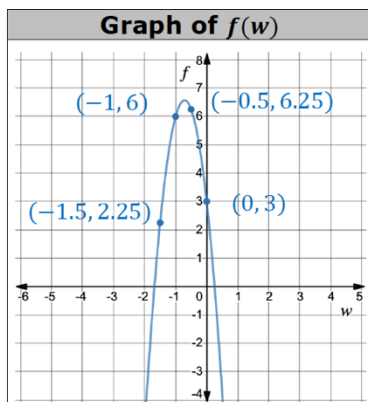


- Were there any areas that caused a calculation error when evaluating?
- Does the value you input change what the function does?



How is the process of evaluating the function different when evaluating for $f(-10)$ versus $f(w) = -45$?

1.2.4 Guided Instruction: Evaluating Functions (continued)



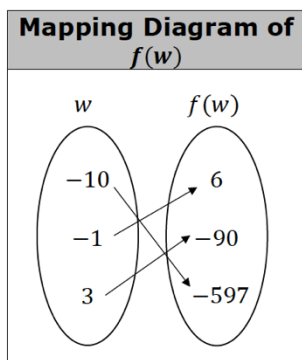
Find each value from the graph.

5. $f(-1.5) = 2.25$

6. $f(0) = 3$



Prior to modeling question 5, review the graph of the function using decreasing and increasing language that students used in the prior activity. This will help reinforce graphing protocols and language. When modelling question 5, connect the input value to the x -value and output value to the y -value.



Find each value from the mapping diagram.

7. $f(3) = -90$

8. Find the value of w when $f(w) = 6$.
 -1



Consider using a foldable to show students how one function can be represented in multiple ways. The top of the foldable can be the overall function and there would be three squares under it, each representing that function being represented in a table, graph, and mapping diagram form.

1.2.5 Try It: Function Notation

Component Overview: Students will practice evaluating functions provided in multiple forms. For this activity, provide students with instructions to complete each question on their own and check their work with a partner. After completing the activity consider bringing all students back together to review the answers and steps for each question.



1.2.5 Try It: Function Notation

1. Given the function $c(y) = -6y - 5$, find each value.

a. $c(-4) =$
19

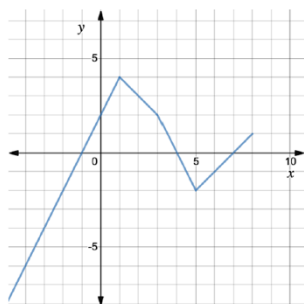
b. $c\left(\frac{3}{4}\right) =$
-9.5

c. For what value of y is $c(y) = 9$?
-2.3

2. Given $g(k) = 2k^2 - 4$, find $g(-5)$.
46

3. Given $h(x) = \sqrt{2.5x + 1}$, find $h(3.2)$.
3

4. Use the graph of $y = f(x)$ below to find each value.



a. $f(1) =$ **4**

b. $f(-2) =$ **-2**

c. For what value of x is $f(x) = -4$? **-3**



Student performance on these questions can provide data for what students are retaining from this lesson, and what additional supports may be necessary from prior learning deficits. Possible scenarios:

- Unable to input the correct value into the correct location in the function
- Calculation errors
- Unable to identify a solution using a graph



Watch for student answers to question 1 part c and question 4 part c. Many students who do well with evaluating functions mistake these types of questions where it isn't an input and evaluate scenario. If students are missing these two but are getting the others correctly, consider reviewing this concept with students again.



Allow students to share out their answer for each question. They should share their work to illustrate how they arrived at the final answer.

1.2.6 Your Turn!

Component Overview: Students will work independently to practice evaluating functions given different forms. For this activity, students should be released to complete questions 1 – 3 on their own. Call students back together to review the answers for each question.



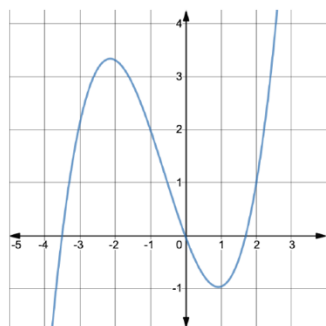
1.2.6 Your Turn!

1. Find each value given the function $m(t) = \frac{t}{3} + 6$.

a. $m(-9) =$
3

- b. Find the input for which $m(t) = 13$.
21

2. Find each value using the graph of the function $f(x)$.



a. $f(-3.5) =$ **0**

b. $f(2) =$ **1**

c. Find x when $f(x) = 4$.
2.5

3. Use the table of values representing $D(r)$ to find each value.

a. $D(-4) =$ **9**

b. $D(-6) =$ **9**

r	$D(r)$
-6	9
0	-5
-4	9
2	0
9	7

- c. Is $D(r)$ a function? Explain why or why not.
Yes, $D(r)$ is a function. For each input there is one unique output.



Consider using the time for this activity to provide differentiated support to students who may be struggling with the lesson content.

Use the information related to student performance from the prior activity, such as where they input the value and what they calculate, to determine which students may need to be pulled together for small group review.

1.2.7 Wrap-Up: Ticket Out the Door

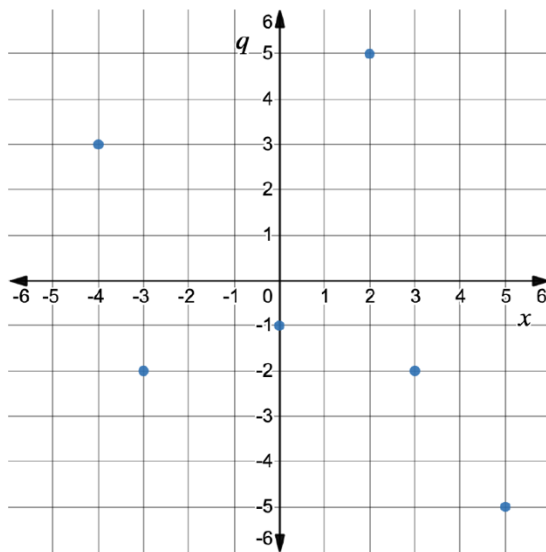
Component Overview: Students will complete a Ticket Out the Door that assesses their ability to evaluate functions. Each student will independently do this activity. The teachers will be able to get accurate data about the student's understanding of the lesson content.



1.2.7 Wrap-Up: Ticket Out the Door

- Given the function $h(t) = t^3 - 2t$, find $h(-7)$.
 $h(-7) = -329$

- Use the graph of $q(x)$ to find $q(-3)$.



$q(-3) = -2$



Function notation is used throughout the course. Use the information from this Wrap-Up to know how well your students understand evaluating functions algebraically and from a graph. As students continue to evaluate using functions throughout the course, it is more important that you look for the student to progress than to have full mastery at the end of this lesson.

For students who do not perform well on this Wrap-Up activity, determine if there is a part of the concept they may not understand. For example, students who get question 1 incorrect but question 2 correct may understand the concept of input values but may have calculation issues. Then, use this information to inform your teaching.